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How Small Changes Can Influence Your Vote: Interactive Dashboard of the Danish Kandidattest

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Abstract

Voters increasingly rely on Voting Advice Applications (VAAs) as tools to compare their own beliefs to political candidates prior to an election. One such VAA is the "Kandidattest" created by a major Danish news outlet, Politiken. A previous analysis by researchers [1] found the algorithm of this "Kandidattest" to be very sensitive to minor change, leading to instability in the computed results. This report expands on the prior investigation by introducing an interactive simulation tool allowing for real-time adjustment to algorithmic weight change, random question removal, and the modification of user responses. By testing those adjustments, the data demonstrates that arbitrary changes can drastically alter a user's top political match. The dashboard offers researchers valuable insights into a system that influences the voting decision of millions across Denmark.

1. Introduction

Algorithms have become an integral part of every person's life, influencing everything from social media content feeds to crucial political decisions. In Denmark, more than half of the population uses the "Kandidattest", a Voting Advice Application (VAA) created by major news outlet Politiken, to help voters compare their policy preferences with those of political candidates. While users trust these systems to provide an objective ranking, a previous investigation [1] shows that the results are strongly dependent on hidden algorithmic weighing choices. With undecided voters and voters with low levels of interest in politics being more likely to follow the advice provided by the VAA [2], there exists a high risk of algorithmic bias, influencing the elections.

This report presents a new interactive simulation tool, allowing users to change

the algorithm's settings in real-time. By manipulating the values that determine how the algorithm computes the match scores, users can observe how slight adjustments alter the final ranking. The dashboard also lets the user drop random questions to show how removing a topic can impact the results.

The dashboard can be accessed [here](#).

2. Methodology: How the Kandidattest works

The "Kandidattest" is a tool that quantifies policy alignment between voters and political candidates. The system uses a mathematical formula to convert users' answers into agreement percentage.

2.1. Survey Structure

The VAA consists of a 20 question survey created by journalist to represent a broad spectrum of policy issues such as climate change, economy and education. For each question, the participant responds with:

- Likert Scale: A score answering the question on an ordered scale from 1 indicating "Completely Disagree" to 5 denoting "Completely Agree".
- Importance Level: Users can further mark the question as Important, Neutral or Not Important to reflect how strong their concerns are about a given issue.

2.2. The Matching Logic

The algorithm determines the match score by calculating a distance score (d_i) for each question, that represents the absolute difference between the user's answer (u_i) and the political candidates answer (p_i). The distances range from 0 (perfect agreement) to 4 (completely opposed views). In order to aggregate the individual scores for each question, the system utilizes a weighted

lookup. The weights (w) are assigned based on both the distance (d_i) and the selected importance level (I_i) by the user:

Importance	Dist. 0	Dist. 1	Dist. 2	Dist. 3	Dist. 4
Not Important	12	15	18	21	24
Neutral	6	12	18	24	30
Important	0	9	18	27	36

Weight lookup table

2.3. Final Agreement Calculation

The system sums the weights across 20 questions in the dataset to create the total weighted distance, which can range from 0 to 720. The calculation is presented by the formula:

$$\text{Total weighted distance} = \sum_{i=1}^q w(d_i, I_i)$$

To make the results interpretable, the algorithm normalizes the sum by converting it into a percentage. Firstly, the disagreement percentage is calculated by dividing the total weighted distance by the maximum possible distance of 720 and multiplying it by 100. Then the resulting score is subtracted from 100 to output the agreement percentage, the final result shown to the user.

$$\text{Disagreement \%} = 100 \times \left(\frac{\text{Total weighted distance}}{\text{Max weighted distance (720)}} \right) \quad (1)$$

$$\text{Agreement \%} = 100 - \text{Disagreement \%} \quad (2)$$

3. Dashboard functionality

The dashboard allows the user to interact with the matching system in real-time and observe the resulting changes. The tool contains responses collected from actual political candidates of all parties from Copenhagen during the 2022 general elections to

ensure the simulation reflects the actual political landscape of Denmark. The created dashboard contains four primary fields.

Weight Matrix Settings: This section allows you to control the weights that the algorithm assigns to the responses of the questions. By editing these values, you decide how heavily the system penalizes the difference between your answers and the candidates. The matrix has been designed to imitate the original weight lookup table and make it simple to edit the assigned numbers manually. It contains three controls: Randomize Weight to try arbitrary scenarios, Reset To Standard Gradient to remove the aspect of importance and Reset To Start Defaults to return to original configuration used by "Kandidattest". The results of the computation are immediately visible in the Match Results section with the newly applied changes.

Robustness Test: This feature tests the robustness of the system by randomly dropping a number of questions selected by the user. It reveals how dependent the results are on a specific questions. By sliding the Questions To Remove bar, you can observe the changes on the leaderboard based on the amount of questions dropped.

Match Results: This table displays the live output of the algorithm. It ranks candidates from 12 main political parties based on the agreement percentage. The display ranks the top matches, allowing you to see instantly how changes in weights or question count alter the recommendations. The top match is highlighted in red.

Political Response Matrix: This matrix visualizes the actual responses of Copenhagen political candidates from the 2022 election. It maps the answers of all 12 major Danish parties with your own Users Answers. By using a heatmap, this section shows exactly where you align or differ with different parties. You can edit your responses to the questions and adjust the importance score as you see fit. It also contains the Randomize User Profile feature,

for easier profile testing.

4. Design Choices

The website employs series of aesthetic choices to increase the affordance of the program, making the usage more intuitive. By a mix of features such as consistent color palette that uses different hue to contrast highlights and varying text sizes, the dashboard establishes a clear visual hierarchy.

4.1. Interactive Controls

The simulation contains two main tools for testing the VAA algorithm, the weight matrix and the robustness test sections. They are presented in Figure 1 and Figure 2 respectively.

The weight matrix, copies the original lookup table used in the studied VAA with the names of the columns being in a slightly darker shade for easier distinction. It also contain three buttons of different colors, each chosen to suggest the unique purpose of its action and to show that they can be interacted with. Under the matrix there is a legend describing the meaning of the columns with the names of the scores made red for contrast.

The random question dropper is a simple slider with a straight forward caption and counter on the right, enabling the user to intuitively choose the amount of question wanted to be removed from the equation. The section is even further highlighted with orange background to denote that the panel is interactive.

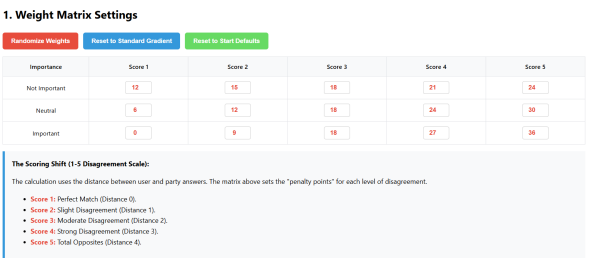


Figure 1: The interactive control panel allowing for real-time weight adjustments.

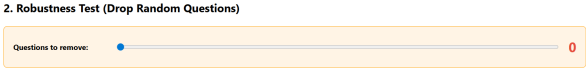


Figure 2: The slider, allowing to drop a chosen amount of questions.

4.2. Match Results Leaderboard

The Match Result section allows for an easy comparison of the real-time updated ranking of matching parties. The results are shown side by side ordered from left to right for a simple assessment of the outcome of changes. The party with the highest agreement metric is further highlighted with red background to indicate the best match. The Figure 3 shows an example of how it looks.

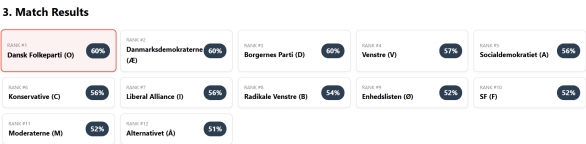


Figure 3: The Match Result section, displaying top political parties ranked based on the agreement metric.

4.3. Political Response Section

The Political Response Matrix includes answers to all 20 questions provided by a political candidate from each party as it can be seen in Figure 4. The responses range from 1 to 5 with the hue of the background changing from white (1) to dark blue (5). This method enables an easy visual comparison of the answers by a quick check. The user's scores are on the top of the matrix and are separated from the politician's responses using a black line to help with distinction and show the interactive element. Furthermore, the section includes the option to adjust the importance of each question with simple drop down button that limits the possible answers. The matrix also includes a Randomize User Profile feature with a green button that stands out from the rest of the visualization.

4. Political Response Matrix (12 Parties)

Kandidat User Profile

Entity	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
User Answers	2	5	1	1	1	5	1	1	5	1	1	5	5	5	5	2	5	5	5	5
Importance	None	None	None	None	None	None	None	None	None	None	None	None	None	None	None	None	None	None	None	None
Enhedslisten (E)	4	5	5	5	2	1	1	1	2	4	5	5	4	2	4	2	1	2	1	1
Alternativet (A)	1	5	5	5	2	1	1	1	5	5	5	4	5	1	2	1	2	1	1	1
SP (S)	5	5	4	5	5	2	4	2	1	5	4	5	2	2	5	1	5	5	1	1
Socialdemokratiet (S)	5	4	4	5	4	4	4	2	2	5	4	4	4	2	5	1	4	4	1	2
Radikale Venstre (R)	4	4	5	5	2	1	4	5	1	2	5	5	1	5	4	1	4	5	1	1
Moderaterne (M)	2	1	4	5	2	1	5	4	5	5	4	5	2	2	4	1	4	4	1	5
Venstre (V)	1	2	2	5	4	4	5	4	4	1	5	4	2	5	5	1	5	5	2	4

Figure 4: The political response matrix, displaying answers to all the question by the user and each party.

5. Conclusion and Discussion

With the popularization of algorithms such as VAAs, it is important to ensure that the tool guiding decisions of millions remains stable and transparent. This report and the interactive dashboard highlight that the "Kandidattest" is not an objective and neutral tool for calculating alignment of political views but a program sensitive to changes due to the underlying technical choices. The investigation shows that when developers select weights for the algorithm and choose particular questions, they have the power to steer the political direction of many voters. The visual representation helps overcome the issue of transparency by transforming the VAA algorithm into a visible and understandable process.

AI Usage

Large Language Models (LLMs) were utilized throughout the study as a supportive tool for script development

References

1. Astante, G., Sinatra, R., & Sekara, V. *It Takes So Little to Change So Much: Investigating the Robustness of a Danish Voting Advice Algorithm*. Networks, Data and Society (NERDS), IT University of Copenhagen, 2025.
2. Tromborg, M. W., & Albertsen, A.

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